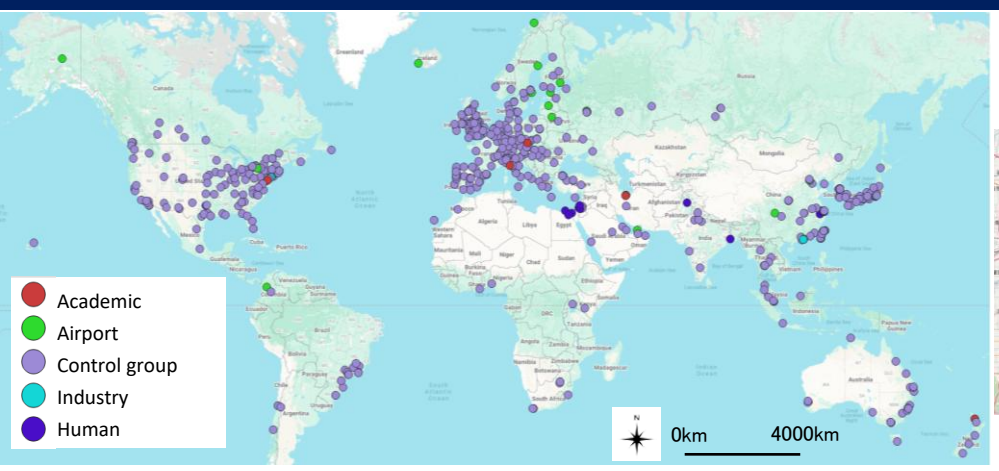
Impact of Campus Accessibility on University Performance and Regional Economy

キャンパスのアクセス性が大学のパフォーマンスと
地域経済に与える影響

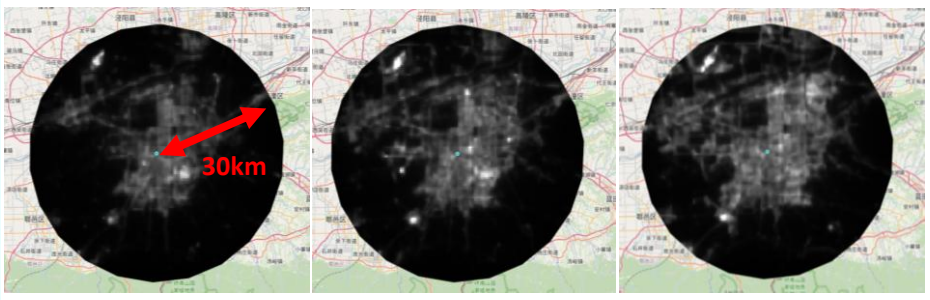
Sayaka Oshima
大島 彩也夏



2015: NTL 9.7
Research Quality 3.5

2019: NTL 15.3
Research Quality 40.2

2024: NTL 18.6
Research Quality 76.3



Northwestern Polytechnical University, China

1 Background

Positive impact of university on local economy

- Universities generate human capital, innovation, and regional economic growth.

Importance of University Campus Accessibility

- Campus accessibility enhances research activity through increased collaboration.

Research question

Do improvements in campus accessibility accelerate university performance and regional revitalization?

2 Goals

This study aims to:

1) Measure campus accessibility globally;
2) Identify its causal impact on university performance;
3) Examine its impact on regional economic activity

Definition of Campus Accessibility : access to airports, other universities (academic), population, and industries

3 Methods

Empirical Analysis 1
Difference-in-differences analysis
Campus accessibility → University performance

Empirical Analysis 2
Difference-in-differences analysis
University performance → Regional economy

Discussion on policy implications

Assumptions in Causal Analysis

- University performance = THE performance scores (Teaching, Research Environment, Research Quality, Industry, and International Outlook)
- Regional economy = VIIRS nighttime lights

Data

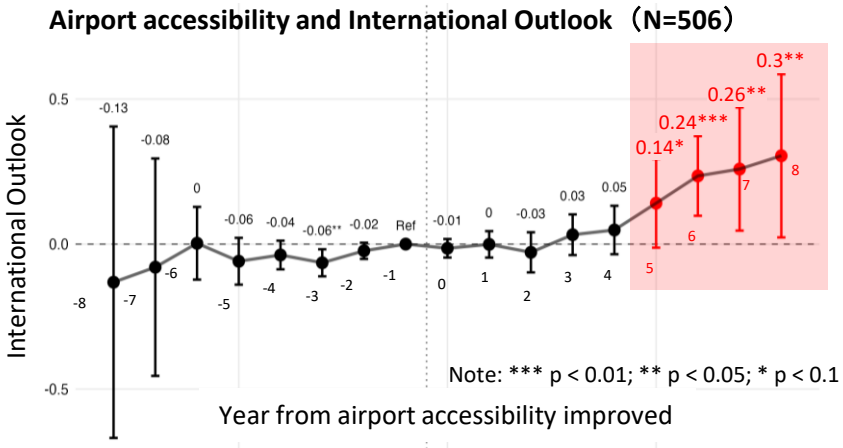
- Global panel of 691 universities (2016–26, THE rankings)
- VIIRS nighttime lights
- Population (UN DESA) and business sector (Forbes Global 2000)

4 Results

Impacts of Accessibility on Performance

- Accessibility improved university performance.

Airport accessibility and International Outlook（N=506）



Note: *** p < 0.01; ** p < 0.05; * p < 0.1

Airport accessibility:
positive impacts on
- Research Quality
- International Outlook

Academic accessibility:
positive impacts on
- Industry
- Research Environment

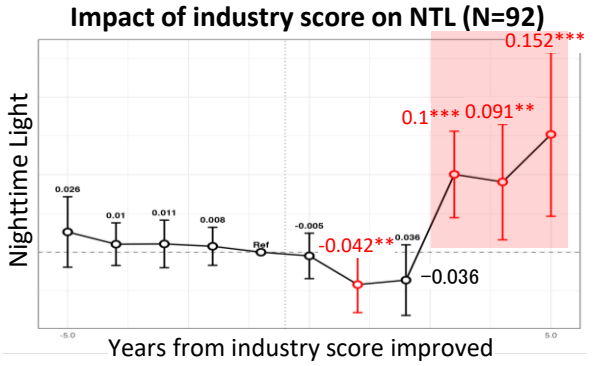
- Airport/Academic accessibility improved Teaching scores in Asia while improved research score in other region.

Impacts of University Performance on Economy

- Industry performance improved regional economy.

Industry Score of university:
positive impacts to regional economy
(after ~3 years)

Impact of industry score on NTL (N=92)



Note: *** p < 0.01; ** p < 0.05; * p < 0.1

5 Conclusions

- Campus accessibility (especially airport, academic access) is a strategic investment for economy.
- Integrating university–industry access into policy promotes knowledge clusters.

Sayaka Oshima
Final Study
of undergraduate
program in 2026
Supervisor: Prof. Hironori Kato

